

Syarfandi "Andi" Achmad

Cloud / SRE Engineer | Cloud Architecture & Infrastructure Specialist Makassar, Indonesia | +62 852-5652-2335 | syarfandi.achmad@gmail.com | Portfolio: syarfandi.github.io | [LinkedIn](#)

Professional Summary

Dynamic **Cloud Engineer** with 4+ years of experience in architecting high-performance **cloud and on-premise server** ecosystems and scalable infrastructure. Expert at designing and implementing robust cloud solutions using **AWS, GCP, and Kubernetes**. Dedicated to delivering operational excellence through automated **Terraform** workflows and cloud-native optimizations. A proactive leader in **asynchronous remote teams**, focused on driving digital transformation and maximizing cloud ROI for global organizations.

Core Technical Competencies

- Infrastructure & Orchestration:** Kubernetes (K8s), Terraform (IaC), Docker, Helm, AWS/GCP, Vercel/Netlify/Firebase, Linux System Administration.
- Cloud Strategy & FinOps:** Cloud Migration, Multi-cloud Architecture, Cost Optimization (FinOps), Cloud Governance, PaaS/SaaS Integration.
- CI/CD & Automation:** GitHub Actions, GitLab CI, Jenkins, PM2, Python/Bash Scripting.
- Data & Middleware:** Supabase, PostgreSQL (Optimization & Scaling), RESTful API Architecture, Redis.
- Soft Skills:** Asynchronous Collaboration, Technical Writing, Remote Team Leadership, Strategic Planning.

Technical Projects

- Kawan Sedarah (Blood Stock):** A **command center** dashboard that tracks **blood stock** availability across donor units in **real-time**, powered by **Supabase** as its core backend.
- Portal Data Sulsel:** A unified information system delivering **data visualization, monitoring**, and religious **statistical analysis** across all districts in South Sulawesi in **real-time**.
- AI Virtual Assistant:** A **Conversational AI**-powered virtual assistant featuring **Realtime Lipsync** on 3D avatars, backed by **Voice AI** and **LLM** for immersive conversational interactions.
- GetKasir Enterprise POS:** A large-scale **Enterprise POS** ecosystem with **100+ modules**, integrating **AI-driven sales prediction**, **biometric authentication**, and a scalable **cloud-native** architecture.
- PPID Kemenag Kota Parepare:** A public information disclosure portal for **PPID Kemenag** providing access to **official documents, performance reports**, and an intelligent chatbot powered by **Google Gemini AI**.
- OP Tracker:** A premium logbook for tracking **One Piece** watch progress without login — equipped with automatic **filler filter, saga & arc** navigation map, and a gamified **pirate rank** system.
- QR Scanner Presence App:** A modern **attendance solution** leveraging **QR Code** scanning for instant participant check-in, built on a **Next.js + Supabase** stack with **real-time** sync.
- Malmora (Reseller & Dropship):** A **reseller & dropship** marketplace tailored for Muslim lifestyle products, enabling anyone to sell without physical stock across **Sulawesi and Eastern Indonesia**.
- Email Server Makassar Gov:** A city-scale **Mailcow**-based email cluster serving **200+ official domains** for Makassar City Government, hardened with **SPF/DKIM/DMARC** security protocols and **high-availability** setup.
- Helpdesk Kota Makassar:** An **e-government** system for handling **citizen complaints** digitally — featuring **real-time ticket tracking**, automated escalation, and cross-department integration across **Makassar City agencies**.
- Virtual Museum Makassar:** An immersive digital museum experience with **360° panoramic** tours, interactive **3D artifact** collections, and **audio-visual** narratives preserving South Sulawesi cultural heritage.
- Al-Haram VR:** A high-precision **virtual reality** simulation of **Al-Masjid Al-Haram** and **Al-Masjid An-Nabawi**, developed in **Unity** and optimized for both **VR headsets** and mobile devices as a Hajj & Umrah educational tool.
- Paralluta (Lalu Lintas):** A smart platform for **traffic management** in Makassar City — covering **road condition monitoring**, violation logging, and centralized **field officer coordination**.
- SOP Pelaporan Insiden Siber:** A **SOP** framework and **cyber incident response protocols** designed for **Makassar City Government** IT infrastructure, aligned with **BSSN** and **NIST Cybersecurity Framework** standards.

Professional Experience

Cloud Infrastructure Lead (Civil Servant) | Ministry of Religious Affairs (Regional Office)

2025 – Present | Makassar, Indonesia

- Spearheaded** the digital transformation of regional religious services, migrating legacy systems to modern cloud architectures.
- Orchestrated** the regional data infrastructure (**Data Sulsel**), enabling real-time statistical analysis for over 9 million residents.
- Strengthened** organizational security posture by implementing automated compliance checks and standardized incident response protocols.
- Automated** internal reporting workflows, reducing manual data processing time by 60% through custom integration scripts.

Cloud Solution Architect | Department of Communication & Informatics (Diskominfo)

2023 – 2025 | Makassar, Indonesia

- Managed** complex on-prem server environments and infrastructure for over **200+ department domains (Satuan Kerja)** across the city government.
- Engineered** a secure and resilient government email cluster using **Mailcow**, ensuring 99.99% availability and robust anti-spam/security filtering.
- Secured** government digital assets by leading the technical response team during the **BSSN National Cyber Exercise**, maintaining 100% uptime during simulated attacks.

Co-Founder | GetKasir (Enterprise POS Startup)

2019 – Present | Remote/Project-based

- Scaled** the cloud infrastructure for a massive **100+ module enterprise retail ecosystem**.
- Implemented** automated cloud backup and recovery strategies, ensuring 99.9% data durability.
- Optimized** PostgreSQL database clusters and Supabase instances to handle high-concurrency retail transactions during peak periods.
- Standardized** technical documentation for infrastructure scaling and disaster recovery, enabling seamless asynchronous collaboration.

SQA & Cloud Infrastructure Engineer | Malmora (Startup)

2020 – 2023 | Remote

- Automated** full-stack testing frameworks and CI/CD pipelines for a high-scale **Mobile Marketplace Reseller** platform.
- Streamlined** mobile app release cycles, ensuring rapid delivery of features while maintaining rigorous quality standards.
- Orchestrated** cloud staging and production environments, reducing infrastructure costs by 25% through resource optimization and serverless utilization.

Secretary | Nurul Huda Mosque

2021 – 2026 | Makassar, Indonesia

- Managed** organizational administration and coordinated the 2021-2026 board of directors.
- Optimized** community asset management and digital documentation for the mosque's operations.

Certifications & Awards

- National Cyber Exercise #10** – BSSN (2024)
- Top 10 Winner** – Samsung Indonesia Next Apps 4.0 Gear VR Challenge (2017)
- Microsoft CityApp Appathon** – Participant & Developer

Education

- Bachelor of Computer Science (S.Kom)** | UIN Alauddin Makassar *GPA: 3.22 / 4.00*

Cloud Expertise Keywords

AWS, Google Cloud Platform (GCP), Azure, Terraform, CloudFormation, Kubernetes, Docker, Serverless, Lambda, Cloud Run, IAM, VPC, Load Balancing, CDN, Route53, S3, Cloud Storage, Observability, CloudWatch, Stackdriver, Infrastructure as Code (IaC), High Availability, Disaster Recovery, Cloud Security, FinOps.

Future-Ready Technical Roadmap

To stay competitive in the global DevOps & Infrastructure market, I am actively implementing:

- Zero Trust Architecture:** Identity-aware proxies and 'Never Trust, Always Verify' principles.
- eBPF Security:** Deep kernel-level observability and real-time security monitoring.
- GitOps:** Using ArgoCD/Flux for automated and declarative infrastructure synchronization.
- Serverless Security:** Securing AWS Lambda/Edge functions and automated IAM least-privilege auditing.